Project Proposal: Multilingual Safety Across Machine Translation

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We propose to study how reliably LLM safety protections transfer across languages. Specifically, we will evaluate whether prompts that are refused in English can produce harmful responses when translated into low-resource languages, such as Swahili, and whether translating harmful responses back into English changes their safety classification. The subjectivity in translating languages creates a challenge for models to generalize prompts appropriately, regardless of the language they are in.

When AI jailbreak evaluations are written, they often focus on prompts in high-resource languages such as English, meaning users of less resourced languages may encounter different safety protections. This problem is challenging because not only does translation not perfectly preserve semantics, but also failures of safety, translation, and evaluation are tangled.

We will source different types of prompts from several datasets. In order to ensure the model can detect malicious intent, **JailBreakBench**, which consists of 200 distinct benign and misuse behaviors, will allow us to confirm both the benchmark and fine-tuned models can detect malicious intent. Datasets such as **XQuAD** can provide multilingual prompts which will verify the model does not over-refuse safe requests as it learns to reject harmful ones. Finally, multilingual datasets such as **MultiJail**, **XSafety**, and **LinguaSafe** will be used to evaluate cross-lingual blind spots.

There are several algorithms available for us to build upon. Deng, Zhang, Pan & Bing (2024) developed an attack methodology using MultiJail by feeding harmful queries in various resource languages into models. Wang et al (2024) provide an evaluation framework by breaking down 14 different harm categories across parallel prompts across different languages. Finally, Yong et al. (2023) propose an annotation schema utilizing three-way labeling as a clear output from the model: BYPASS, REJECT, or UNCLEAR. By combining these methodologies, we will evaluate the performance of a pretrained multilingual LLM such as Llama 3.1 or Qwen 2.5 on the prompts provided by our datasets across three languages; English, a translated medium-resource language, and a translated low-resource language.

References

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